

Profile: Lead AI Engineer bridging the gap between theoretical machine learning and physical lab environments. 5+ years of experience designing scalable, cloud-based AI systems to accelerate scientific discovery. Expert in orchestrating self-driving labs, optimizing in-silico data pipelines, and deploying predictive models (GNNs, PINNs) to solve complex search problems in the molecular sciences.

Skills & links: Python, Rust, C#, TypeScript, PyTorch, Tensorflow, AWS, GCP, Docker, SQL, REST, websockets, French, Spanish.

Lead AI Engineer (Aug 2026 - Present) | Senior AI Engineer (Nov 2025 - Aug 2026) | AI Engineer (Apr 2024 - Nov 2025) — [Materiom](#)

Funded by Google, leading the technical development and implementation of Materiom's natural-polymer discovery stack.

- **ML models:** GNNs, PINNs, and tabular models for materials property prediction & inverse design from experimental & in-silico datasets. Achieving a ~10% error in unseen materials.
- **Closed-loop discovery:** development of Active Learning & Bayesian Optimization workflows for model development and materials discovery. Achieving at least 50% sample efficiency over conventional methods.
- **Self-driving Lab:** design, development & orchestration of Materiom's robotics platform, reaching throughput of 10s of characterized materials per day, including molecular spectroscopy, as well as end material properties.
- **In-silico & experimental data pipelines:** Scaled data assets 500x by developing large-scale CG-MD simulation pipelines utilizing Active Learning, generating 100s of highly efficient samples per day.
- **[Materiom AI](#):** Multi-Agent system with access to 100k+ materials and various search APIs, exposed publicly, serving 10000+ user sessions across 130 countries since its deployment, and 10+ enterprise partnerships.

AI Engineer (Sep 2021 - Apr 2024) — [dRISK](#) (absorbed by [Nexar Inc.](#))

Led the design, development, and deployment of dRISKs Autonomous Vehicle testing, training & validation platform:

- **Scenario-based adversarial validation platform:** developed pipelines for validating RL control systems, LiDAR, radar, and camera based perception stacks, specifically validating PPO, CNN & transformer-based architectures under high-risk scenarios.
- Led technical delivery of customer-facing projects for major clients, including **Wayve**, **NVIDIA**, **Luminar**, and **TATA Motors**.

AI Researcher / Master's Thesis (Jun 2021 - Sep 2021) — [Neurolabs](#) & The University of Edinburgh

Thesis: *Graph Convolutional Neural Networks for Synthetic Data Generation using Unity* (Awarded Distinction). Researched and developed geometric deep learning pipelines to generate high-fidelity synthetic spatial data. Read more [here](#).

Education & Scholarships

MSc in Artificial Intelligence ([The University of Edinburgh](#), grade: distinction), MSc in Mechatronic Engineering ([Arts et Métiers ParisTech](#), grade: 17/20), BSc in Product Design Engineering ([EAFIT University](#), grade: 4.75/5.0). [Colfuturo Excellence Scholarship](#), [Eiffel Excellence Scholarship](#).

Patents & Applied Research

- **Material discoveries undergoing patentability reviews:** Engineered the AI discovery stack at Materiom that produced 5+ breakthrough natural-polymer candidates currently undergoing patenting and commercial trialing.
- ([Application No. PCT/US2024/032276](#)): ML algorithm using graph theory and A* search to augment trajectory data for training autonomous vehicle RL models.

Talks

AI for Materials Science, University of the Arts Central Saint Martins, [A day in the life of an AI Startup](#), EAFIT University, [Synthetic Data Generation for ML](#), Machine Learning Colombia